

§2.3: De Jong–van Straten’s theory: Milnor fibers from Germ Deformations

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August 23, 2026

1. MAIN RESULT FROM DJvS98

As stated from the Plamenevskaya & Starkston paper, the main result is: *deformations of the sandwiched singularity can be encoded via deformations of the germ $(C,0)$ satisfying certain hypotheses.* §2.3 explores a special case relevant to the paper.

2. NOTATION

Definition 2.1. Let C be a curve with irreducible components (curvetas) C_i . For each component C_i , we associate a weight $w_i \in \mathbb{Z}_{>0}$ corresponding to w_i marked points concentrated at $0 \in C_i$.

Definition 2.2. In the context of the algebraic framework used by Plamenevskaya and Starkston, a subscheme captures not only the geometric support (the underlying set of points) but also infinitesimal data and multiplicities.

Definition 2.3. A *picture deformation* is a 1-parameter deformation of the curve germ $(C, 0)$ together with a deformation of the zero-dimensional subscheme w .

A picture deformation acts as a transformation that breaks down complex curve singularities into smooth, simple components. As the parameter s moves away from 0, the tangled branches of a curve (curvetas) unfold into smooth pieces that intersect only transversally. During this process, concentrated marked points at the origin scatter along the branches, with all intersections between different branches aligned to occur at these moving points.

A picture deformation resolves a complex curve singularity by unfolding its branches (curvetas) into a reduced, smooth state for $s \neq 0$, where all intersections are transverse. The deformation requires that the moving decoration (marked points) w_s on the normalization $\widetilde{C}^s$ contains the multiplicity scheme m_s of intersections, ensuring that all crossings are accounted for as distinct, marked points.

Definition 2.4. A *picture deformation* $\mathcal{C}^S$ of the decorated germ $(\mathcal{C}, w)$ with smooth components $C_1, \dots, C_m$ over a germ of smooth curve $(S, 0)$ is given by a δ -constant deformation $\mathcal{C}^S \rightarrow S$ of $\mathcal{C}$ and a flat deformation $w^S \subset \widetilde{\mathcal{C}^S} = \widetilde{C} \times S$ of the scheme w , such that for $s \neq 0$, the divisor w^s is reduced, the only singularities of $\mathcal{C}^s$ are ordinary multiple points, and $m^s \subset w^s$.

When $s \neq 0$, you can stop overthinking the abstract algebra and just visualize w^s as a normal set of points $\{p_1, p_2, \dots, p_n\}$ sitting right on the smooth curves. These points do two jobs: some are placed exactly where the curves cross $(C_i^s \cap C_j^s)$, and the rest are just “free points” hanging out on a single branch away from the mess.

The big takeaway from de Jong and van Straten (Theorem 1.3) is that you can build 2D smooth surfaces out of these 1D curve drawings. Every picture deformation gives you a smoothing of a surface singularity $(X, 0)$, and every possible smoothing comes from one of these pictures.

To actually build the smooth manifold (the Milnor fiber), the recipe is straightforward: you take $\mathbb{C}^2$ with your curve branches on it, blow up every single marked point (both the crossings and the free ones), and then throw away the curves themselves. Whatever space is left over is your Milnor fiber. This is huge because it means you can study complicated surface smoothings just by drawing and tweaking these curve pictures.

2.1. Remark 2.8 (in paper)

Remark 2.5. To make this construction compact and keep track of boundaries, we look at what happens inside a closed Milnor ball $B \subset \mathbb{C}^2$. For a small deformation ($s \neq 0$), the curves $\mathcal{C}^s$ exit the boundary $\partial B = S^3$ cleanly, and all the marked points p_i stay safely inside the interior of B .

If we blow up B at these marked points to get $\tilde{B}$, the messy intersecting curves untangle completely, turning into a collection of separate, smooth disks. To find the compact 4-manifold version of our Milnor fiber W , we just take $\tilde{B}$ and scoop out thin, separate tubes T_i around each of these smooth disks. After smoothing out the sharp edges/corners, this leaves us with the compact manifold $W = \tilde{B} \setminus \bigcup_{i=1}^m T_i$, whose Stein structure naturally matches up with the complex structure from the blow-up.

REFERENCES

- [1] R. E. Gompf and A. I. Stipsicz, *4-Manifolds and Kirby Calculus*, Graduate Studies in Mathematics, vol. 20, American Mathematical Society, Providence, RI, 1999.
- [2] O. Plamenevskaya and L. Starkston, *Unexpected Stein fillings, rational surface singularities, and plane curve arrangements*, *Geometry & Topology*, 27(3), 2023, pp. 1083–1166. Preprint available at arXiv:2006.06631 [math.GT], <https://arxiv.org/abs/2006.06631>.
- [3] G.-M. Greuel, C. Lossen, and E. Shustin, *Introduction to Singularities and Deformations*, Springer Monographs in Mathematics, Springer-Verlag, Berlin, 2007.
- [4] T. de Jong and D. van Straten, *Deformation theory of sandwiched singularities*, *Duke Mathematical Journal*, 95(3), 1998, pp. 451–522.